

Chapter 9: Aortic Stenosis: Diagnosis, Treatment and its Impact on Healthcare in India

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Abstract

Aortic stenosis is a heart condition that primarily affects older adults. The supply of blood from the heart to other organs is restricted due to improper opening and closing of the aortic valve cusps and calcium deposition in the aorta and left ventricle. The patient's survival rate increases if the disease is detected early and preventive actions are taken. Besides the availability of different diagnostic tools for detecting aortic stenosis, echocardiography is considered the gold standard to evaluate the clinical parameters and visualize the structure of the heart and valve. Based on the severity and risk factors, different intervention methods can be employed, including surgical aortic valve replacement (SAVR) and transcatheter aortic valve replacement (TAVR). However, balloon valvuloplasty and transcatheter aortic valve replacement are the two most significant treatments to cure severe AS. These intervention methods have a significant impact in India. Furthermore, early diagnosis and intervention strategies for aortic stenosis have substantially affected patient outcomes and quality of life in India. This chapter explores the prevalence, causative factors, diagnostic mechanisms, and treatment of aortic stenosis within the Indian context.

Keywords: Aortic Stenosis (AS), Mitral Stenosis (MS), surgical aortic valve replacement (SAVR), TAVR (Transcatheter Aortic Valve Replacement)

1 Introduction

Aortic stenosis (AS) is a kind of valvular heart disease caused by stiffening of the aortic valve leaflet due to calcium deposition in the left ventricle and aorta or congenital issues (Ito & Oh, 2022). In Western countries, aortic stenosis (AS) is the second most prevalent valvular heart disease, affecting 2% of people aged below 75 years and 6% of those above 75 years (Mantovani et al., 2021). In India, degenerative AS is very common among older people. Younger subgroups are associated with bicuspid aortic valve compared to those described in Western cohorts (A. Sahu et al., 2020). From the Indian cohort, it was estimated that 58.1% of patients with AS have degenerative tricuspid aortic valve disease, followed by bicuspid aortic valve at 25.0%, with rheumatic disease uncommon at 2.9% (A.K. Sahu et al., 2020). The burden of AS is rising with the growing prevalence of diabetes, hypertension, and other risk factors (Gupta et al., 2020).

- 1.1 What is Aortic Stenosis:** When the aortic valve obstructs blood flow from the heart to other organs, mainly due to structural abnormalities and calcification of the leaflets in the aortic valve (Maharani et al., 2025). It can cause symptoms like breathlessness, tiredness, and irregular heartbeat. The anatomical structure of the heart is represented in Fig 1.

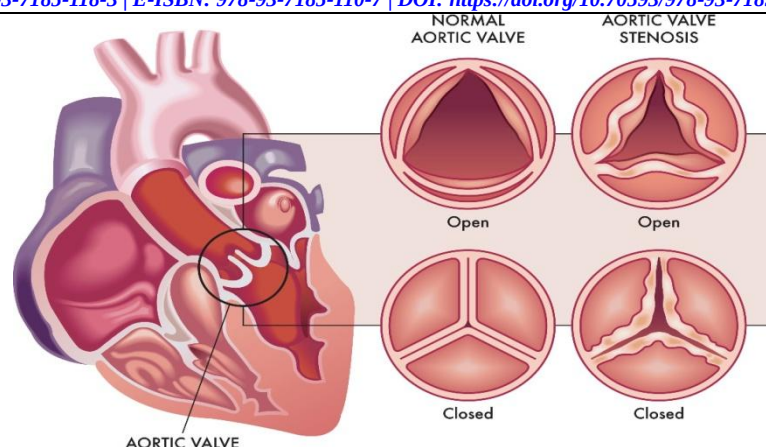


Fig 1: Anatomy of a normal Aortic valve and one with stenosis (source: <https://avicennacardiology.com/blog/what-is-aortic-stenosis/>)

1.2 Types of Aortic stenosis: Depending on the causing factors, AS can be classified into the following types:

- **Degenerative Aortic Stenosis:** In India and other developed countries, degenerative aortic stenosis (AS) is prevalent among the older population. It is a condition which gradually forming calcium in the aortic valve and left ventricle, restricting the normal flow of blood through the valve.
- **Congenital Aortic Stenosis:** The most common cause of congenital AS is defects in the leaflets or the absence of leaflets from birth. The valve leaflets are opened to transmit oxygenated blood to the left ventricle; this is called systole. The valve leaflets are closed to prevent blood from flowing back.
- **Rheumatic Aortic Stenosis:** In developed countries, rheumatic AS is very common, whererheumatic fever is the main factorthat damages the aortic valve.
- **Pathological Aortic Stenosis:** It occurs due toother heart conditions or diseases like cardiomyopathy, hypertension and connective tissue disorders, which can evolve into this type of disease.

1.3 The Burden of AS in India: In India, AS is significant for many reasons.

- **Increasing burden of cardiovascular disease:** Cardiovascular diseases (CVDs) are among the non-communicable diseases replacing communicable diseases in India's epidemiological transition.
- **Growing Ageing Population:** India's rapid growth of the ageing population has resulted in a significant rise in the number of older adults. As a result, it increases the number of age-related diseases such as aortic stenosis (Olasińska-Wisniewska et al., 2013).
- **Impact on Healthcare Infrastructure:** Aortic stenosis poses a significant economic impact as itrequires expensive interventions or surgical treatments, such as SAVR (Thaden et al., 2014).
- **Access to Healthcare:** The availability of healthcare services varies across different regions in India. People in rural areasoften suffer due to a lack of healthcare and treatment (Verma & Dash,

2020). Research on aortic stenosis enables us to identify healthcare issues and learn how to better identify and treat those who are affected early on.

2 Literature Review

Aortic stenosis (AS) is a critical cardiac disease where accurate and early diagnosis is vital for effective treatments. The patient undergoes different medical assessments, including clinical evaluation, physical examination and imaging techniques to accurately diagnose aortic stenosis (Zamorano et al., 2015); (Billig et al., 2024). Echocardiography and Doppler echocardiography are considered the main diagnostic tools for determining the severity of AS by quantifying the different indicators, such as valve areas using the continuity equation (Zamorano et al., 2015); (Nistri et al., 2008). Another important diagnostic method, computed tomography (CT) imaging, is employed to measure the calcification of the aortic valve and plan for transcatheter aortic valve implantation (TAVI) (Yoon et al., 2018; Pflederer & Achenbach, 2010). Even though there are various diagnostic tools, AS can't be detected early; sometimes it is identified after death (Nistri et al., 2008). This happens because AS can be sometimes asymptomatic that don't show any symptoms in the patient like fasting of heart beat and chest pain (Nistri et al., 2008). Furthermore, CT imaging may result in larger aortic valve areas than echocardiography; this should be taken into account for AS diagnosis (Pflederer & Achenbach, 2010). In summary, the diagnosis of AS is a rigorous process which not only examines clinical parameters but also utilizes the various imaging modalities, with echocardiography of diagnosis. The precise diagnosis and management of AS is dependent on expert analysis with the integration of all available data. It is crucial to understand how underdiagnosis and limitations of each diagnostic method may affect people and increase the mortality rate (Nistri et al., 2008; Pflederer & Achenbach, 2010; Zamorano et al., 2015).

The prevalence of aortic stenosis depends on factors such as age, geographical region, and patients' existing health issues. Generally, the condition is more common among older people with structural changes in the heart valve (Otto et al., 2021). In developed countries like the United States and Europe, with increasing life expectancy, the number of cases of aortic stenosis is also increasing. According to the study (Brennan et al., 2012), approximately 2% of Indian people suffered from severe aortic stenosis. There is no evidence to conduct a study of the demographics of AS in the Indian context; the study is solely based on hospital data. Among Indian patients of isolated aortic stenosis, the age distribution was bimodal, with peaks noted in childhood and the 50-59 years age group, reflecting congenital and acquired disease (Sahu et al., 2020). Among acquired isolated aortic stenosis, 58.1% was degenerative tricuspid valve disease, and 25.0% was calcified bicuspid valve disease, while rheumatic isolated aortic stenosis was rare at 2.9% (Sahu et al., 2020). Untreated AS leads to a poor prognosis of individual having a median survival of less than 2 years for those with heart failure symptoms (Otto et al., 2021).

To overcome the problem of misdiagnosis with an increasing burden on the healthcare and economic system, a reliable and efficient approach is needed to accurately identify aortic stenosis at an earlier stage. Many researchers have worked on the detection of AS by employing machine learning and deep learning approaches on healthcare data, including imaging and clinical data. In the study (Wessler et al., 2023), a deep learning algorithm was employed on echocardiographic video data, resulting in AUROC values between 0.91 and 0.98. This indicates a strong diagnostic model ability and interpretability over human experts. Similarly, for detecting moderate to severe AS, a deep

learning model based on ECG data has shown outstanding results, with 78% sensitivity, indicating the model's ability over conventional detection methods (Cohen-Shelly et al., 2021). Despite the performance of these models, there are some challenges, including a lack of public data availability, bias in the training data, and data imbalance. Further research is essential to overcome this problem in order to develop a more robust model by applying multimodal data collected from heterogeneous sources for early detection and severity assessment of AS.

3 Diagnostic Tools: A complete diagnosis process involves a combination of physical examination, clinical evaluation and diagnostic tests for assessing the structural abnormalities and other factors affecting AS. Primary diagnostic methods are grouped into two categories. The various diagnostic tools used for AS detection are shown in Fig 2.

3.1 Invasive: These techniques involve entry into the human body by inserting instruments such as surgical biopsies. The methods include cardiac catheterization and biopsy.

- **Cardiac Catheterization:** A catheter or instrument is placed into a blood vessel to visualize the structural images of the heart. This test enables assessment of the severity and provides follow-up treatments.

3.2 Non-invasive: These techniques involve methods that assess the body's internal conditions without penetrating the skin or surgically entering the body. Several noninvasive diagnostic methods are available, including ECG, MRI, Echocardiography, and X-ray.

- **Electrocardiogram (ECG):** This test is used to measure the heart rate of a patient with electrical signals.
- **Echocardiography:** Echocardiography is the most efficient and gold standard method for identifying cardiac diseases, such as aortic stenosis (AS). It can assess the functional and structural aspects of the heart, such as heart chamber size, valve diameter, and cusp movement during systole, as well as peak aortic velocity, mean pressure gradient and aortic valve area. Based on these parameter values, the clinician prepares the diagnosis report and grades the severity of aortic stenosis. Echocardiography was performed from different perspectives. Common echocardiographic views include:
 - **Parasternal long-axis view (PLAX):** This view enables the clinician to visualize and evaluate the size and wall thickness of the left ventricle. It provides a clear visualization of the aortic and mitral valves for identifying abnormalities in them.
 - **Parasternal short-axis view (PSAX):** This standard echocardiographic view shows a cross-sectional view of the heart.
 - **Apical 2-Chamber View:** This is standard for identifying the left ventricle and the left atrium.
 - **Apical 3-Chamber View:** This enables assessment of the left part of the heart, including the aortic valve, left ventricle outflow tract (LVOT) and mitral valve.

- **Apical 4-Chamber View:** This view allows measurement of the wall thickness and motion of the left ventricle.

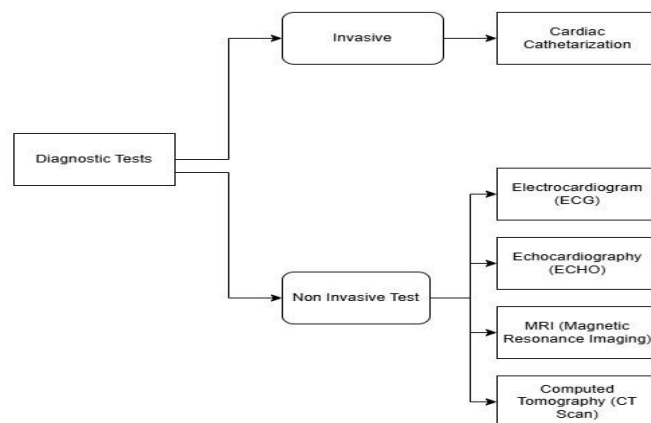


Fig 2: Diagnosis Test used for Aortic Stenosis Detection

4 Treatments

Aortic valve replacement (AVR) is the main approach for patients with severe aortic stenosis, which is surgically performed by transcatheter procedures (Ossola et al., 2026). Since 1960, with the improvement in operative techniques such as surgical AVR, it has been successfully executed which are contributing to reducing mortality rates. For patients at high risk for surgical AVR due to some factors like age, other health-related issues, transcatheter aortic valve replacement (TAVR) has been implemented to treat the patient (Akahori et al., 2018a). Transcatheter aortic valve replacement (TAVR) has been recognized as the best treatment option for high-risk patients who cannot undergo surgical AVR (Sehatzadeh et al., 2013). However, pharmacotherapy has not been effective in halting the progression of AS. It makes AVR the most reliable treatment for severe AS. Further studies are required to develop novel pharmacotherapeutic strategies to slow the progression of AS (Akahori et al., 2018b).

5 Impact of AI in Diagnosis and Management of Aortic Stenosis:

The impact of artificial intelligence on diagnosis and management of aortic stenosis (AS) focuses on screening and assessing the severity via echocardiography. However, AI is evolving for disease prognosis, risk-based referral, and treatment planning (Zhang et al., 2024). Research on AS-based machine learning is grouped into three areas, including early-stage screening, severity/progression assessment, and aiding management decisions. These studies address the issues related to bias, heterogeneity, interpretability, and lack of prospective workflow validation (Chen et al., 2025; Popat et al., 2024). The recent AI-based method demonstrates strong accuracy in the detection of moderate to severe AS on imaging or ECG data from a large external validation cohort (Holste et al., 2023). Modalities include echocardiography video-input models with AUROC 0.94-0.98 and

electrocardiogram (ECG) testing with AUC of 0.85 (Cohen-Shelly et al., 2021). Automating echocardiographic parameters commonly used for AS determination has been validated and performs comparably to human readers for peak velocity, mean gradient, and valve area calculation (Krishna et al., 2023). Recent models have achieved high accuracy for disease severity detection that are validated internally and externally (Park et al., 2025).

Despite the variability in performance between data sources and clinical settings, the feasibility and scalability of AI-based methods have been demonstrated using digital stethoscopes, limited 2D echocardiography (echo), and multimodal screening tools (Ghanayim et al., 2022; Wessler et al., 2023). An automated AS detection method using ECG signal data was collected from 258,607 patients, which enable to identify moderate to severe AS with 78% sensitivity and 74% specificity in one of the largest validation studies. Additionally, an AI model with higher false-positive cases as compared to false negative have long term risks, suggesting that the AI model is detecting a signal associated with latent disease rather than pure noise (Cohen-Shelly et al., 2021). In a recent meta-analysis, sensitivity and specificity pooled across 10 separate studies were 0.83 and 0.81, respectively, with a pooled AUC of 0.909. This evidence supports AI-based AS detection is a scalable approach that can be applied to different signal-based tools such as ECG (Popat et al., 2024). In the study (Ghanayim et al., 2022), the performance of the AI algorithm identifying moderate or severe AS was highest in the external validation cohorts for digital stethoscope models, with a sensitivity of 84 to 90%. The performance of the AI model is better in patients with severe AS compared to those with only mild disease. Automated screening models using only a single view of 2D echo imagery achieved AUROC ranging from 0.91 to 0.96 using limited parasternal long-axis images (Wessler et al., 2023). In particular, AI models had lower accuracy using the A4C view (Elkouahy et al., 2025).

Further studies support applying robust deep learning methods with multi-view imaging data of echocardiography for disease screening and severity assessment. However, model performance was inconsistent between ultrasound views.

6 Conclusions:

Aortic stenosis has a significant impact in India, with challenges in the healthcare system, morbidity and mortality rates, and also affecting the economic condition. Early diagnosis and effective management of AS are possible by a combination of clinical assessment and advanced imaging diagnosis tools such as Echocardiography. Treatment options have evolved, with surgical interventions being the primary choice for severe cases and less invasive procedures, such as TAVR, offering hope for high-risk patients. With the increased burden of aortic stenosis in India, it substantially needed the awareness among people, healthcare improvement, and continued research to enhance diagnostic and treatment methods. To overcome the above challenges, AI with integration into the healthcare system can be effective in improving diagnosis and treatment processes by analyzing medical images more accurately and quickly than radiologists. The process of diagnosis is operator-dependent, and it is a subjective procedure; sometimes there exists bias due to these factors. To overcome these problem and AI based methods are useful in detecting the disease with higher accuracy, precision and recall. This can lead to earlier detection of disease and help clinicians to make more informed decisions about treatment plans (Popat et al., 2024). Further, the accuracy of

early AS detection could also be enhanced by employing a robust AI model by integrating clinical data with different imaging modalities like echocardiography and ECG.

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